

# Aritra Basu

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📍 Sunnyvale, CA

## PROFILE

Senior **systems software engineer** specializing in high-performance networking and user-space data planes. 6+ years of industry experience building **SASE** and **SD-WAN** solutions and open-source networking infrastructure. Active upstream contributor to multiple open-source projects, including **VPP**, **CalicoVPP**, **curl**, and **systemd**.

## SKILLS

|                    |                                                                            |
|--------------------|----------------------------------------------------------------------------|
| <b>Languages</b>   | C, C++, Go, Python, Bash                                                   |
| <b>Networking</b>  | IPv4/IPv6, TCP, UDP, TLS/DTLS, QUIC, HTTP/3, MASQUE, IPsec, BGP, DHCP, DNS |
| <b>Data Plane</b>  | VPP (HostStack), DPDK, eBPF, netfilter                                     |
| <b>Debugging</b>   | gdb, tcpdump, Wireshark, iperf, strace, Valgrind, iptables, Frida          |
| <b>Infra/Tools</b> | Git, Docker, Kubernetes, Ginkgo, Pytest, Jenkins, Ansible, Terraform       |

## EXPERIENCE

**Cisco Systems, Inc.**  
*Software Engineer IV*  
*Software Engineer III*

📍 San Jose, CA  
May 2026 - Present  
Jun 2024 - May 2026

- Member of the **VPP** team within Cisco Security; enhancing **HostStack** - the user-space L4-L7 protocol stack in VPP.
- Advancing VPP dataplane integration (IPv6 support) for **CalicoVPP** CNI in **Kubernetes** pod-networking environment.
- Developing **Go syscall interception** with **Frida** as an alternative to LD\_PRELOAD for Go apps integrating with VPP.
- Implemented **HTTP/3 proxy CONNECT** and **MASQUE CONNECT-UDP** support for **ngtcp2 QUIC** in **curl**.
- Building **MCP** servers, reusable skills, and agentic development / debugging **workflows** for the VPP ecosystem.

**Member of Technical Staff II**  
*VeloCloud, VMware (acquired by Broadcom)*

Jul 2022 - Jun 2024  
📍 Palo Alto, CA

- Active member of the **gateway** and **routing** pillars of the **SASE SD-WAN** data plane and control plane teams.
- Resolved critical technical debt in **BGP route-map** design and worked on a feature to add **OSPFv3** support.
- Implemented **persistent storage for statistical counters** to improve field debugging and operational visibility.
- Contributed to addition of **Mellanox DPDK PMD** support to improve ConnectX NIC performance.

**Product Development Intern**  
*VeloCloud, VMware*

Jun 2021 - Sep 2021  
📍 Palo Alto, CA

- Automated scale-testbed provisioning to enable developer performance testing.
- Configured L2/L3 VLANs, VRFs, and BGP associations across lab infrastructure.
- Containerized edge services to simulate large-scale tunnel deployments.

**Member of Technical Staff I**  
*Netskope*

Jul 2018 - Jul 2020  
📍 Bengaluru, India

- Enhanced the **IPsec** solution with tunnel-status monitoring and per-tunnel bandwidth visibility.
- Added support for multiple **IPsec** tunnels originating from a single source IP.
- Improved the **DTLS** tunnel performance and resolved live production networking and performance issues.
- Migrated the **IPsec** stack from **strongSwan** v5.3.5 to v5.7.2 to remediate security vulnerabilities.

**Software Developer Intern**  
*Netskope*

Jan 2018 - Jun 2018  
📍 Bengaluru, India

- Implemented device fingerprinting to uniquely identify devices steering traffic to the Netskope cloud.
- Enhanced a library for loading Netskope private root CA certificate in Firefox **certdb**.
- Migrated the Netskope Client codebase from Visual Studio 2012 to Visual Studio 2015.

## OPEN SOURCE CONTRIBUTIONS

- curl** - Implemented **HTTP/3 proxy CONNECT** and **MASQUE CONNECT-UDP** support for **ngtcp2 QUIC**.
- VPP** - Contributed to **HostStack** L4-L7 protocol enhancements and **IPv6 (ND Proxy)** improvements.

- **CalicoVPP** - Added **IPv6** support to the VPP data plane integration for **Kubernetes** CNI pod networking.
- **systemd** - Contributed bugfixes (**DHCPv6**) and feature enhancements (**ARP Proxy**) to **networkd**.

## EDUCATION

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### University of California, San Diego (UC San Diego)

*M.S. in Computer Science* (GPA: 3.77/4.0)

Distributed Computing, Operating Systems, Virtualization, Computer Security, Computer Architecture, Design of Algorithms

San Diego, CA

Jan 2021 - Jun 2022

### Vellore Institute of Technology (VIT Vellore)

*B.Tech. in Electronics & Communication Engineering* (GPA: 9.64/10; Silver Medalist)

Computer Communication, Digital Communication, Information Theory, Wireless Communication, Data Structures, VLSI

Vellore, India

Jul 2014 - Apr 2018

## ACADEMIC PROJECTS

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- **Router Design** - Built a software router supporting **ARP/ICMP** processing, **TCP/UDP** packet forwarding, **ARP cache** management, and longest-prefix matching (**LPM**) **routing**.
- **Sandboxing libopus in Firefox** - Used the **RLBox** sandboxing framework to isolate the **Opus** codec in Firefox with minimal audio performance overhead; measured and documented the performance impact.
- **Tribbler (Distributed Key-Value Store)** - Designed and implemented a **distributed, fault-tolerant key-value store in Rust**; benchmarked three data replication strategies at scale based on the Distributed Hash Table (DHT) protocol.

## TEACHING

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### Graduate Teaching Assistant - UC San Diego

Jan 2021 - Jun 2022

- CSE 123 - Computer Networks (Spring 2022, Winter 2022, Spring 2021) - Prof. Aaron Schulman & Prof. Alex Snoeren
- CSE 124 - Networked Services (Fall 2021) - Prof. George Porter
- CSE 141 - Computer Architecture (Winter 2021) - Prof. Bryan Chin